cuRAMSES III: Non-Standard Cosmological Models — CPL Dark Energy, Massive Neutrinos, SIDM, MOND, $f(R)$, nDGP, Symmetron, Dilaton, Galileon, Coupled Dark Energy, and Early Dark Energy in Adaptive Mesh Refinement Simulations

Juhan Kim[1]⋆
[1] *Center for Advanced Computation, Korea Institute for Advanced Study, 85 Hoegiro, Dongdaemun-gu, Seoul 02455, Republic of Korea*

Accepted XXX. Received YYY; in original form ZZZ

ABSTRACT

We present extensions to the cuRAMSES adaptive mesh refinement (AMR) code that enable cosmological hydrodynamic simulations with eleven classes of non-standard physics beyond ΛCDM. The first group operates within the Poisson solver framework: (i) CPL dark energy with dynamical equation of state $w(a) = w_0 + w_a(1-a)$ and density perturbations via a table-based linear response method for arbitrary sound speed c_s^2; (ii) massive neutrino linear response using precomputed CAMB transfer function ratios $R_\nu(k, a)$. The second group addresses dark matter phenomenology: (iii) self-interacting dark matter (SIDM) Monte Carlo scattering with constant, velocity-dependent, anisotropic, and inelastic cross-sections; (iv) Modified Newtonian Dynamics in QUMOND and AQUAL formulations with external field effect (EFE) support. The third group implements scalar field modified gravity theories via Newton–Gauss–Seidel relaxation on AMR levels: (v) $f(R)$ Hu–Sawicki gravity with chameleon screening; (vi) nDGP braneworld gravity with Vainshtein screening; (vii) Symmetron gravity with spontaneous symmetry breaking; (viii) environmentally dependent Dilaton gravity; (ix) cubic Galileon gravity reusing the Vainshtein operator. Finally, two additional modules modify forces and background cosmology: (x) Coupled Dark Energy with an enhanced effective gravitational constant $G_{\rm eff} = G(1 + 2\beta_{\rm cde}^2)$ for dark matter; and (xi) Early Dark Energy in the Doran–Robbers parameterisation, which modifies the Friedmann equation at high redshift. All scalar field models (f(R), nDGP, MOND, Symmetron, Dilaton, Galileon) are mutually exclusive, while coupled DE and early DE can combine freely with any model. Each module is controlled by dedicated namelist blocks and is verified to recover ΛCDM results when disabled. We describe the algorithms, screening mechanisms, unit conversions, and numerical validation, providing a self-contained reference for non-standard cosmological simulations with AMR codes.

Key words: methods: numerical – cosmology: simulations – dark energy – neutrinos – dark matter – gravitation – modified gravity – $f(R)$ gravity – braneworld models

1 INTRODUCTION

The ΛCDM concordance model, with a cosmological constant Λ and cold, collisionless dark matter, provides an excellent fit to a wide range of cosmological observations (Planck Collaboration 2020). Yet several fundamental questions remain open. The nature of the accelerated expansion is unknown: the cosmological constant may be replaced by a dynamical dark energy field whose equation of state evolves with redshift (Chevallier & Polarski 2001; Linder 2003). Neutrino oscillation experiments have established non-zero neutrino masses (Lesgourgues & Pastor 2006; Lattanzi & Gerbino 2017), which suppress the growth of structure on small scales.

On the dark matter side, persistent small-scale tensions — the core–cusp, missing satellites, and too-big-to-fail problems — have motivated self-interacting dark matter (SIDM) models in which DM particles scatter elastically or inelastically with non-negligible cross-sections (Spergel & Steinhardt 2000; Tulin & Yu 2018). An alternative explanation of these small-scale phenomena, as well as of galaxy rotation curves without dark matter, is provided by Modified Newtonian Dynamics (MOND; Milgrom 1983), in which the gravitational acceleration is enhanced below a critical threshold $a_0 \approx 1.2 \times 10^{-8}\,{\rm cm\,s^{-2}}$. The quasi-linear formulation of MOND (QUMOND; Milgrom 2010) recasts this modification as a modified Poisson equation that can be solved with standard numerical methods.

Beyond these phenomenological approaches, a rich land-

⋆ E-mail: kjhan@kias.re.kr

© 2026 The Authors

scape of gravitational theories has emerged that modify general relativity (GR) on cosmological scales. $f(R)$ gravity (Hu & Sawicki 2007; Sotiriou & Faraoni 2010) introduces a scalar degree of freedom through a nonlinear function of the Ricci scalar, with the Hu–Sawicki model providing a chameleon screening mechanism that recovers GR in high-density environments. The Dvali–Gabadadze–Porrati (DGP) braneworld model (Dvali et al. 2000) explains cosmic acceleration through gravitational leakage into an extra dimension; the normal branch (nDGP) exhibits Vainshtein screening (Vainshtein 1972; Nicolis et al. 2009), whereby nonlinear derivative interactions suppress the scalar force near massive sources. The Symmetron mechanism (Hinterbichler & Khoury 2010; Hinterbichler et al. 2011) employs a scalar field with a spontaneous symmetry-breaking potential whose vacuum expectation value (VEV) vanishes in high-density regions, effectively decoupling the fifth force. The environmentally dependent Dilaton model (Brax et al. 2010, 2011) achieves a similar screening via the Damour–Polyakov mechanism in which the coupling function has a minimum at the cosmological field value. Cubic Galileon gravity (Nicolis et al. 2009; Deffayet et al. 2009a,b) features shift-symmetric scalar field interactions with second-order equations of motion, screened by the Vainshtein mechanism.

Several N-body codes have implemented subsets of these models, notably ECOSMOG (Li et al. 2012, 2013), ISIS (Llinares et al. 2014), MG-GADGET (Puchwein et al. 2013), and the RAMSES-based solver of Llinares et al. (2014). The Newton–Gauss–Seidel (N-GS) relaxation method with red–black ordering has proven robust for the nonlinear scalar field equations arising in all these theories (Li et al. 2012; Llinares et al. 2014).

Beyond modifications to gravity, the dark energy sector itself may couple directly to dark matter. Coupled dark energy (CDE) models (Amendola 2000, 2004; Pettorino & Baccigalupi 2008) introduce an interaction term in the dark sector Lagrangian, yielding an effective gravitational constant $G_{\rm eff} = G(1 + 2\beta^2_{\rm cde})$ for DM particles. Independently, the coincidence problem and Hubble tension have motivated Early Dark Energy (EDE) models (Doran & Robbers 2006; Poulin et al. 2019; Hill et al. 2020; Smith & Poulin 2020) in which a non-negligible dark energy component exists at early times ($z \sim 3000$) before decaying rapidly, modifying the expansion history and sound horizon.

Precision cosmological surveys such as DESI (DESI Collaboration 2024), Euclid (Euclid Collaboration 2024), and the Vera C. Rubin Observatory LSST (Ivezić et al. 2019) will constrain these extensions at the per-cent level, demanding equally precise theoretical predictions. Full hydrodynamic simulations that simultaneously capture baryonic physics and non-standard dark-sector models are essential for this programme.

In this paper we describe the implementation of eleven non-standard cosmological modules in CURAMSES (Kim 2026), a suite of algorithmic and GPU-acceleration optimizations for the RAMSES AMR code (Teyssier 2002):

(i) **CPL dark energy perturbations** (Section 3): a table-based linear response method for the Chevallier–Poloncki–Linder (CPL) parameterisation.

(ii) **Massive neutrino linear response** (Section 4): scale-dependent Poisson equation corrections using precomputed CAMB (Lewis et al. 2000) transfer function ratios.

(iii) **Self-interacting dark matter** (Section 5): cell-based Monte Carlo scattering with constant, velocity-dependent, anisotropic, and inelastic cross-sections.

(iv) **QUMOND and AQUAL** (Section 6): Modified Newtonian Dynamics with three operational modes and external field effect support.

(v) $f(R)$ **Hu–Sawicki gravity** (Section 7): chameleon-screened scalar field solved via Newton–Gauss–Seidel relaxation.

(vi) **nDGP braneworld gravity** (Section 8): Vainshtein-screened scalar field with the nonlinear Laplacian operator.

(vii) **Symmetron gravity** (Section 9): spontaneous symmetry-breaking screening with environment-dependent VEV.

(viii) **Dilaton gravity** (Section 10): Damour–Polyakov environmentally dependent coupling.

(ix) **Cubic Galileon gravity** (Section 11): shift-symmetric scalar field with Vainshtein screening, reusing the nDGP operator.

(x) **Coupled dark energy** (Section 12): enhanced gravitational constant $G_{\rm eff} = G(1 + 2\beta^2_{\rm cde})$ for dark matter.

(xi) **Early dark energy** (Section 13): Doran–Robbers parameterisation modifying the background Friedmann equation.

All scalar field models (items 4–9) share a common infrastructure: the `scalar_gr` array for the field value and the N-GS red–black relaxation solver with OpenMP parallelism. Exactly one scalar field model may be active at a time, while coupled DE and early DE combine freely with any other model. All modules compose with the existing CURAMSES infrastructure — k-section domain decomposition, Morton key octree, FFTW3 direct Poisson solver, and OpenMP+MPI hybrid parallelism — without requiring modifications to the core AMR data structures.

Table 1 summarizes the inter-model compatibility. Each module operates at a distinct physical level — background expansion, Poisson source, gravitational force, or particle dynamics — so most combinations are allowed. The sole restriction is that the six scalar field / modified gravity models (MOND, $f(R)$, nDGP, Symmetron, Dilaton, Galileon) are *mutually exclusive* because they share the `scalar_gr` storage array and solve fundamentally different field equations.

The structure of this paper is as follows. Section 2 reviews the unified Poisson solver framework. Sections 3–6 describe the dark energy, neutrino, SIDM, and MOND modules. Sections 7–11 present the five scalar field modified gravity solvers. Sections 12–13 describe the coupled DE and early DE modules. Section 14 presents verification tests, and Section 15 concludes.

2 UNIFIED POISSON SOLVER FRAMEWORK

The cosmological Poisson equation in comoving coordinates reads

$$\nabla^2\phi = \frac{3}{2}\,\Omega_{\rm m}\,a\,\delta_{\rm m}, \tag{1}$$

where ϕ is the gravitational potential, a is the scale factor, $\Omega_{\rm m}$ is the total matter density parameter, and $\delta_{\rm m}$ is the total matter overdensity.

Table 1. Inter-model compatibility matrix for the eleven non-standard modules. ✓ = fully compatible (may be enabled simultaneously), × = mutually exclusive. The six scalar field models (MOND through Galileon) share the `scalar_gr` array and are mutually exclusive; all other combinations are allowed. CPL DE includes its perturbation sub-module. Models are grouped by the physical level at which they operate: background expansion (B), Poisson source (P), force modification (F), or particle dynamics (D).

	CPL DE	Neutrino	SIDM	MOND	$f(R)$	nDGP	Symmetron	Dilaton	Galileon	Coupled DE	Early DE
Level	B/P	P	D	F	F	F	F	F	F	F	B
CPL DE	—	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Neutrino	✓	—	✓	✓	✓	✓	✓	✓	✓	✓	✓
SIDM	✓	✓	—	✓	✓	✓	✓	✓	✓	✓	✓
MOND	✓	✓	✓	—	×	×	×	×	×	✓	✓
$f(R)$	✓	✓	✓	×	—	×	×	×	×	✓	✓
nDGP	✓	✓	✓	×	×	—	×	×	×	✓	✓
Symmetron	✓	✓	✓	×	×	×	—	×	×	✓	✓
Dilaton	✓	✓	✓	×	×	×	×	—	×	✓	✓
Galileon	✓	✓	✓	×	×	×	×	×	—	✓	✓
Coupled DE	✓	✓	✓	✓	✓	✓	✓	✓	✓	—	✓
Early DE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	—

In the standard ΛCDM case, all matter is treated as cold and collisionless: $\delta_{\rm m} = \delta_{\rm cb}$ (CDM + baryons). When massive neutrinos or clustering dark energy are present, the Poisson source must include their density perturbations:

$$\nabla^2\phi = \frac{3}{2}\,a\,[\Omega_{\rm cb}\,\delta_{\rm cb} + \Omega_\nu\,\delta_\nu + \Omega_{\rm DE}(a)\,\delta_{\rm DE}]\,, \tag{2}$$

where $\Omega_{\rm cb} = \Omega_{\rm m} - \Omega_\nu$, and $\Omega_{\rm DE}(a) = \Omega_\Lambda\, f_{\rm DE}(a)\, a^3$ is the physical dark energy density in units of the critical density at the current epoch.

Rather than evolving δ_ν and $\delta_{\rm DE}$ as additional fluid variables, we adopt the *linear response* approach: each non-standard perturbation is expressed as a scale-dependent multiple of the CDM+baryon perturbation,

$$\delta_X(k,a) = R_X(k,a)\,\delta_{\rm cb}(k,a), \tag{3}$$

where $X \in \{\nu, {\rm DE}\}$ and the response function $R_X(k,a)$ is precomputed from CAMB transfer functions. Substituting into equation (2) and transforming to Fourier space, the modified Green's function becomes

$$\hat{\phi}(\mathbf{k}) = G(k)\,\hat{\delta}_{\rm cb}(\mathbf{k}) \times \underbrace{\left[1 + \frac{\Omega_\nu}{\Omega_{\rm cb}}\,R_\nu(k,a)\right]}_{\text{neutrino factor}} \times \underbrace{\left[1 + \frac{\Omega_{\rm DE}(a)}{\Omega_{\rm cb}}\,R_{\rm DE}(k,a)\right]}_{\text{DE factor}}, \tag{4}$$

where $G(k) = -1/k^2$ is the standard Laplacian Green's function (with appropriate normalisation). Each correction factor defaults to unity when the corresponding module is disabled, ensuring bit-wise recovery of ΛCDM.

cuRAMSES employs a two-level Poisson solver (Guillet & Teyssier 2011; Kim 2026): a multigrid (MG) V-cycle for AMR fine levels and an FFT direct solve for the uniform base level. The k-dependent corrections in equation (4) are applied mode-by-mode in the FFT solver, where Fourier modes are explicitly available. For the MG solver, which operates in real space, only k-independent corrections can be applied; these enter through the pre-factor

$$4\pi G_{\rm eff} = \frac{3}{2}\,\Omega_{\rm cb}\,a\,\mathcal{S}, \tag{5}$$

where $\mathcal{S} = 1$ in standard ΛCDM. Table 2 summarises how each module modifies the two solver levels.

Table 2. Summary of Poisson solver modifications. 'FFT' denotes the base-level direct solve; 'MG' denotes the multigrid V-cycle for fine AMR levels.

Module	k-dependent?	FFT path	MG path
Neutrino	Yes	$\times\,[1 + (\Omega_\nu/\Omega_{\rm cb})R_\nu]$	$\Omega_{\rm cb}$ replaces $\Omega_{\rm m}$
DE ($c_{\rm s}^2 = 0$)	No	$\times\,[1 + (\Omega_{\rm DE}/\Omega_{\rm cb})R_{\rm DE}]$	$4\pi G_{\rm eff}$ boosted
DE ($c_{\rm s}^2 > 0$)	Yes	$\times\,[1 + (\Omega_{\rm DE}/\Omega_{\rm cb})R_{\rm DE}]$	— (FFT only)
DE (no table)	Yes	Helmholtz factor	— (FFT only)

3 CPL DARK ENERGY WITH PERTURBATIONS

3.1 Background Evolution

The CPL parameterisation (Chevallier & Polarski 2001; Linder 2003) models the dark energy equation of state as

$$w(a) = w_0 + w_a\,(1-a), \tag{6}$$

where w_0 is the present-day value and w_a captures the time variation. The dark energy density evolves as

$$\frac{\rho_{\rm DE}(a)}{\rho_{\rm DE}(1)} = a^{-3(1+w_0+w_a)}\,\exp[-3w_a\,(1-a)] \equiv f_{\rm DE}(a). \tag{7}$$

For ΛCDM ($w_0 = -1$, $w_a = 0$), $f_{\rm DE}(a) = 1$ identically. The Friedmann equation becomes

$$E^2(a) \equiv \frac{H^2(a)}{H_0^2} = \frac{\Omega_{\rm m}}{a^3} + \frac{\Omega_k}{a^2} + \Omega_\Lambda\,f_{\rm DE}(a), \tag{8}$$

which governs the time-stepping and scale factor evolution in cuRAMSES.

3.2 Table-Based Linear Response

For arbitrary c_s^2, including the important clustering limit $c_s^2 \to 0$, the DE perturbation $\delta_{\rm DE}$ cannot be obtained analytically from $\delta_{\rm cb}$ in a simple closed form. We therefore precompute an effective response ratio $R_{\rm DE}(k,a)$ from CAMB (Lewis et al. 2000) via the *Weyl potential method* and store it in an ASCII table for use at runtime.

A naïve extraction of $\delta_{\rm DE}/\delta_{\rm m}$ from CAMB transfer functions is problematic because CAMB works in the *synchronous* gauge, whereas the N-body code solves the Newtonian-gauge Poisson equation $\nabla^2\phi = 4\pi G\, a^2\, \bar\rho\, \delta$. At low redshift, the synchronous-gauge $\delta_{\rm DE}$ can exceed its Newtonian-gauge counterpart by a factor $\sim$26 for $c_s^2 = 0$, leading to a severe over-correction of the gravitational potential.

Instead, we define an effective response function

$$R_{\rm DE}(k,a) \equiv \left[\frac{\Psi_{\rm W}^{(\rm clust)}(k,a)/\delta_{\rm m}^{(\rm clust)}}{\Psi_{\rm W}^{(\rm smooth)}(k,a)/\delta_{\rm m}^{(\rm smooth)}} - 1\right] \frac{\Omega_{\rm m}}{\Omega_{\rm DE}(a)\, a^3} \tag{9}$$

where $\Psi_{\rm W}$ is the gauge-invariant Weyl potential returned by CAMB, and the superscripts 'clust' and 'smooth' refer to two CAMB runs with $c_s^2 = c_{s,\rm target}^2$ and $c_s^2 = 1$ (non-clustering DE), respectively. This ratio isolates the extra gravitational potential sourced by DE clustering, automatically in the correct (Newtonian) gauge.

The table generation proceeds as follows. We run CAMB twice with the same CPL parameters (w_0, w_a) but different sound speeds: the target c_s^2 and a reference $c_s^2 = 1$. For each of 50 scale factor values a_i (logarithmically spaced from $a = 0.01$ to $a = 1$) and 200 wavenumbers k_j (logarithmically spaced from $k = 10^{-4}$ to $10\,h\,{\rm Mpc}^{-1}$), we extract the Weyl potential and

$$\delta_{\rm m}(k,a) = \frac{\Omega_{\rm cdm}\,\delta_{\rm cdm} + \Omega_{\rm b}\,\delta_{\rm b}}{\Omega_{\rm m}} \tag{10}$$

from both runs, compute $R_{\rm DE}$ via equation (9), and store $R_{\rm DE}(k_j, a_i)$ in an ASCII table.

At runtime, CURAMSES reads this table once at the first Poisson solve. For each Fourier mode with physical wavenumber

$$k_{\rm phys} = \frac{2\pi}{L_{\rm box}}\sqrt{k_x^2 + k_y^2 + k_z^2} \quad [h\,{\rm Mpc}^{-1}], \tag{11}$$

the response function $R_{\rm DE}(k_{\rm phys}, a)$ is obtained by bilinear interpolation in $(\log_{10} k, \log_{10} a)$ space. The DE correction factor in equation (4) is then

$$\mathcal{F}_{\rm DE}(k,a) = 1 + \frac{\Omega_{\rm DE}(a)}{\Omega_{\rm cb}}\, R_{\rm DE}(k,a), \tag{12}$$

where $\Omega_{\rm DE}(a) = \Omega_\Lambda\, f_{\rm DE}(a)\, a^3$.

For the important case $c_s^2 = 0$, dark energy clusters at all scales identically: $R_{\rm DE}$ is independent of k. This allows the MG solver to also capture the DE correction via a real-space boost to the $4\pi G_{\rm eff}$ pre-factor (equation 5):

$$\mathcal{S} = 1 + \frac{\Omega_{\rm DE}(a)}{\Omega_{\rm cb}}\, R_{\rm DE}(k_{\rm ref}, a), \tag{13}$$

evaluated at $k_{\rm ref} = 10^{-3}\,h\,{\rm Mpc}^{-1}$ (any value suffices since $R_{\rm DE}$ is k-independent).

3.3 Quasi-Static Helmholtz Fallback

When no CAMB table is provided and $c_s^2 > 0$, a quasi-static approximation is available. In the sub-horizon, quasi-static limit, the DE perturbation satisfies a Helmholtz equation that can be combined with Poisson's equation to yield a modified Green's function (Bean & Doré 2004; Sapone & Kunz 2010):

$$\mathcal{F}_{\rm DE}(k) = \frac{\tilde k^2 + \kappa^2}{\tilde k^2 + \kappa^2 + \alpha}, \tag{14}$$

where $\tilde k = 2\pi\sqrt{k_x^2 + k_y^2 + k_z^2}$ is the dimensionless Fourier wavenumber, and

$$\kappa^2 = \frac{a^2\, E^2(a)}{c_s^2}\left(\frac{L_{\rm box}}{c/H_0}\right)^2, \tag{15}$$

$$\alpha = \frac{3}{2}\,\Omega_\Lambda\, f_{\rm DE}(a)\,\frac{1 + w(a)}{c_s^2}\left(\frac{L_{\rm box}}{c/H_0}\right)^2. \tag{16}$$

For ΛCDM, $1 + w = 0$ so $\alpha = 0$ and $\mathcal{F}_{\rm DE} = 1$. This approach is valid only for $c_s^2 > 0$; for $c_s^2 \to 0$, both κ^2 and α diverge, and the table-based method must be used instead.

3.4 Namelist Configuration

The CPL dark energy module is activated by setting `de_perturb=.true.` in `&RUN_PARAMS`. The equation of state and sound speed are specified in a dedicated `&CPL_PARAMS` namelist:

```
&CPL_PARAMS
w0 = -0.9
wa = 0.1
cs2_de = 0.0
de_table = 'de_table.dat'
/
```

The code selects the solver path automatically: table-based if `de_table` is provided, quasi-static Helmholtz otherwise (with $c_s^2 > 0$ required).

4 MASSIVE NEUTRINO LINEAR RESPONSE

4.1 Physical Motivation

Massive neutrinos with total mass $\sum m_\nu > 0$ free-stream out of gravitational potential wells on scales below the neutrino free-streaming length, suppressing the matter power spectrum by a fraction $\Delta P/P \approx -8\,\Omega_\nu/\Omega_{\rm m}$ on small scales (Lesgourgues & Pastor 2006). Current constraints from CMB and large-scale structure data give $\sum m_\nu < 0.12\,{\rm eV}$ at 95 per cent confidence (Planck Collaboration 2020), corresponding to $\Omega_\nu \lesssim 0.003$.

The standard approach of representing neutrinos as N-body particles (Brandbyge et al. 2008; Viel et al. 2010) suffers from severe shot noise: the high thermal velocities of neutrinos make them nearly homogeneous, so resolving the small perturbations $\delta_\nu \ll 1$ requires an impractically large number of particles. The linear response method (Ali-Haïmoud & Bird 2013; Bird et al. 2018) avoids this problem entirely by computing δ_ν analytically from the CDM+baryon perturbation.

4.2 Transfer Function Table

We compute the neutrino response function

$$R_\nu(k,a) = \frac{T_\nu(k,a)}{T_{\rm cb}(k,a)} \tag{17}$$

using CAMB, where T_ν and $T_{\rm cb}$ are the neutrino and CDM+baryon transfer functions, respectively. The table spans 200 wavenumbers (10^{-4}–$10\,h\,{\rm Mpc}^{-1}$, log-spaced) and 50 scale factor values (0.01–1.0, log-spaced), and is generated by a Python script `generate_neutrino_table.py` included with CURAMSES.

The default cosmology uses Planck 2018 parameters (Planck Collaboration 2020): $\Omega_{\rm b} = 0.0486$, $\Omega_{\rm m} = 0.3089$, $h = 0.6774$, $\sum m_\nu = 0.06\,{\rm eV}$ (one massive eigenstate). For three degenerate neutrinos with $\sum m_\nu = 0.06\,{\rm eV}$, the density parameter is $\Omega_\nu \approx 0.00140$. At the table boundaries, R_ν ranges from -0.009 to 1.332: neutrinos trace CDM perturbations on large scales ($R_\nu \to 1$) and are suppressed below the free-streaming scale ($R_\nu \to 0$).

4.3 Poisson Solver Integration

4.3.1 FFT base-level solver

At the base level, where the grid is uniform and an FFT direct solve is performed, the neutrino correction is applied mode-by-mode after the forward real-to-complex FFT:

$$\hat{\phi}(\mathbf{k}) = G(k)\,\hat{\delta}_{\rm cb}(\mathbf{k}) \times \left[1 + \frac{\Omega_\nu}{\Omega_{\rm cb}}\,R_\nu(k_{\rm phys},a)\right]. \tag{18}$$

The physical wavenumber $k_{\rm phys}$ is computed via equation (11). This correction is applied in both the distributed (FFTW MPI slab decomposition) and local (`MPI_ALLREDUCE` + serial FFTW) code paths.

4.3.2 Multigrid fine-level solver

The MG solver operates in real space and cannot apply k-dependent corrections directly. Instead, the Poisson source pre-factor is modified from $\Omega_{\rm m}$ to $\Omega_{\rm cb}$:

$$4\pi G_{\rm eff} = \frac{3}{2}\,\Omega_{\rm cb}\,a\,{\rm scale}, \tag{19}$$

where scale is the code-to-physical unit conversion. This correctly accounts for the fact that only CDM+baryon matter contributes to the real-space source term; the neutrino contribution, being k-dependent, is captured exclusively through the FFT base-level correction. The approximation is valid because neutrino perturbations are small ($\delta_\nu \ll \delta_{\rm cb}$ on AMR-refined scales) and the base-level FFT captures the dominant large-scale neutrino effect.

4.4 Namelist Configuration

The neutrino module requires `use_neutrino=.true.` in `&RUN_PARAMS`, plus a `&NEUTRINO_PARAMS` block:

```
&NEUTRINO_PARAMS
omega_nu = 0.00140
neutrino_table = 'neutrino_table.dat'
/
```

At startup, the code validates that $\Omega_\nu > 0$ and the table file exists; otherwise the module is disabled with a warning.

4.5 Consistency with Dark Energy Corrections

When both neutrino and dark energy modules are active, the correction factors compose multiplicatively in equation (4). Each factor is independent and defaults to unity when disabled. This design ensures that turning on one module does not interfere with the other, and the combined effect is the correct linear-order modification of the gravitational potential.

5 SELF-INTERACTING DARK MATTER

5.1 Physical Motivation

In the CDM paradigm, dark matter is collisionless. Self-interacting dark matter (SIDM) models introduce a non-zero scattering cross-section σ/m that allows DM particles to exchange momentum and energy, potentially resolving small-scale tensions with observations (Spergel & Steinhardt 2000; Tulin & Yu 2018).

Astrophysical constraints span several orders of magnitude in relative velocity: dwarf galaxies probe $v \sim 30$–$100\,{\rm km\,s^{-1}}$, Milky Way-mass haloes $v \sim 200\,{\rm km\,s^{-1}}$, and galaxy clusters $v \sim 1000$–$3000\,{\rm km\,s^{-1}}$. A constant cross-section $\sigma/m \sim 1\,{\rm cm^2\,g^{-1}}$ is consistent with dwarf-scale observations but may be too large for clusters (Kaplinghat et al. 2016). Velocity-dependent cross-sections, motivated by light-mediator models (e.g. dark photon exchange), naturally accommodate these constraints (Feng et al. 2009; Tulin et al. 2013; Kaplinghat et al. 2016).

5.2 Monte Carlo Scattering Algorithm

We implement a cell-based Monte Carlo scattering algorithm following the standard approach used in SPH/mesh codes (Rocha et al. 2013; Robertson et al. 2017; Banerjee et al. 2020). The algorithm operates on each AMR leaf cell independently, processing all levels from `levelmin` to `nlevelmax`.

5.2.1 Particle collection and pairing

For each leaf cell (identified by `son(ind_cell)==0`), we collect all DM particles by traversing the particle linked list (`headp` → `nextp`) and selecting particles with `idp` > 0 and `tp` ≤ 0 (the latter condition includes both ground-state and excited-state DM in the inelastic model; see Section 5.5). If the number of DM particles $N_{\rm dm} < N_{\rm min}$ (default 2), the cell is skipped.

The collected particles are randomly shuffled using the Fisher–Yates algorithm (Knuth 1997), which produces a uniformly random permutation in $\mathcal{O}(N_{\rm dm})$ operations. Sequential pairs are then formed: $(1,2)$, $(3,4)$, …, yielding $\lfloor N_{\rm dm}/2 \rfloor$ pairs. If $N_{\rm dm}$ is odd, the last particle is unpaired and does not scatter.

5.2.2 Scattering probability

For each pair (i,j) with masses m_i, m_j and velocity vectors $\mathbf{v}_i$, $\mathbf{v}_j$, the scattering probability is

$$P = \frac{\sigma/m(v)\,\bar{m}\,v_{\rm rel}\,\Delta t}{V_{\rm cell}}\,(N_{\rm dm} - 1), \tag{20}$$

where $v_{\rm rel} = |\mathbf{v}_i - \mathbf{v}_j|$ is the physical relative velocity, $\bar{m} = (m_i + m_j)/2$ is the mean particle mass (in physical units), Δt is the physical timestep, and $V_{\rm cell} = (\Delta x\, \ell/a)^3$ is the physical cell volume with ℓ the code length unit and a the scale factor. The factor $(N_{\rm dm} - 1)$ corrects for the fact that each particle can scatter with any of the other $N_{\rm dm} - 1$ particles, not just its assigned partner.

A scatter event occurs if a uniform random deviate $\mathcal{U}(0,1) < P$. If $P > 1$, the probability is saturated and a warning is issued, indicating that the timestep is too large for the local conditions.

The unit conversions are:

$$\bar{m}_{\rm phys} = \texttt{mp}(i)\, \rho_0\, \ell^3, \tag{21}$$

$$v_{\rm rel}^{\rm phys} = v_{\rm rel}^{\rm code}\, v_0/a, \tag{22}$$

$$V_{\rm cell}^{\rm phys} = (\Delta x\, \ell/a)^3, \tag{23}$$

$$\Delta t_{\rm phys} = \texttt{dtnew}\, t_0, \tag{24}$$

where (ℓ, t_0, ρ_0, v_0) are the RAMSES code-to-CGS conversion factors obtained from the `units()` subroutine.

5.2.3 Velocity update

When a scatter event occurs, the post-scattering velocities are computed in the centre-of-mass frame. The COM velocity is

$$\mathbf{v}_{\rm cm} = \frac{m_i\, \mathbf{v}_i + m_j\, \mathbf{v}_j}{m_i + m_j}, \tag{25}$$

and a new relative velocity direction $\hat{\mathbf{n}}$ is drawn (see Sections 5.4.1 and 5.4.2 for the angular distribution). The new velocities are

$$\mathbf{v}_i' = \mathbf{v}_{\rm cm} + \frac{m_j}{m_i + m_j}\, |v_{\rm rel}'|\, \hat{\mathbf{n}}, \tag{26}$$

$$\mathbf{v}_j' = \mathbf{v}_{\rm cm} - \frac{m_i}{m_i + m_j}\, |v_{\rm rel}'|\, \hat{\mathbf{n}}, \tag{27}$$

where $|v_{\rm rel}'| = |v_{\rm rel}|$ for elastic scattering. This construction guarantees *exact* momentum conservation: $m_i \mathbf{v}_i' + m_j \mathbf{v}_j' = m_i \mathbf{v}_i + m_j \mathbf{v}_j$ by construction, and exact kinetic energy conservation for equal-mass elastic scattering. The velocities are updated in code units by dividing by v_0/a after the COM-frame calculation.

5.3 Velocity-Dependent Cross-Sections

Three cross-section models are supported, selected by the `sidm_type` parameter:

5.3.1 Constant

$$\frac{\sigma}{m} = \sigma_0, \tag{28}$$

where $\sigma_0 \equiv$ `sidm_cross_section` in units of $\mathrm{cm^2\, g^{-1}}$.

5.3.2 Yukawa

Motivated by Yukawa-type dark force models with a light mediator of mass m_ϕ (Feng et al. 2009; Tulin et al. 2013), the transfer cross-section in the Born regime scales as

$$\frac{\sigma(v)}{m} = \frac{\sigma_0}{[1 + (v/v_0)^2]^2}, \tag{29}$$

where v_0 is a reference velocity related to m_ϕ. This produces strong scattering at low velocities (dwarf haloes) and rapid suppression at high velocities (clusters), in agreement with observational constraints (Kaplinghat et al. 2016).

5.3.3 Power-law

A generic power-law dependence:

$$\frac{\sigma(v)}{m} = \sigma_0 \left(\frac{v}{v_0}\right)^n, \tag{30}$$

with a floor at $v = 1\,\mathrm{km\,s^{-1}}$ and a cap at $10^4\,\sigma_0$ to prevent numerical overflow. Common choices include $n = -4$ (Born-regime Yukawa) and $n = -2$.

In all cases, v is the physical relative velocity $v_{\rm rel}$ of the pair in $\mathrm{km\,s^{-1}}$.

5.4 Angular Distributions

5.4.1 Isotropic scattering

The default angular distribution is isotropic: the scattered direction $\hat{\mathbf{n}}$ is drawn uniformly on the unit sphere,

$$\begin{aligned} \cos\theta &= 2\mathcal{R}_1 - 1, \\ \phi &= 2\pi\, \mathcal{R}_2, \\ \hat{\mathbf{n}} &= (\sin\theta\cos\phi,\ \sin\theta\sin\phi,\ \cos\theta), \end{aligned} \tag{31}$$

where $\mathcal{R}_1, \mathcal{R}_2 \sim \mathcal{U}(0,1)$.

5.4.2 Anisotropic (Rutherford-like) scattering

For light-mediator models, the differential cross-section is forward-peaked. We implement a regularised Rutherford distribution (Kahlhoefer et al. 2014; Robertson et al. 2017):

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} \propto \frac{1}{(1 - \cos\theta + 2\varepsilon)^2}, \tag{32}$$

where ε is a regularisation parameter controlling the degree of forward peaking ($\varepsilon \to 0$: strongly forward-peaked; $\varepsilon \to \infty$: isotropic). The scattering angle is drawn by CDF inversion:

$$\cos\theta = 1 + 2\varepsilon - \frac{1}{b + \mathcal{R}(a - b)}, \tag{33}$$

where $a = 1/(2\varepsilon)$ and $b = 1/(2 + 2\varepsilon)$. The azimuthal angle ϕ is uniform on $[0, 2\pi)$.

The scattered direction in the lab frame is obtained by constructing an orthonormal basis $(\hat{\mathbf{v}}_{\rm rel}, \hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2)$ aligned with the incident relative velocity and rotating $\hat{\mathbf{v}}_{\rm rel}$ by (θ, ϕ) in this basis.

5.5 Inelastic Scattering (iSIDM)

We implement a two-state inelastic SIDM model following Tucker Smith & Weiner (2001) and Vogelsberger et al. (2019). Dark matter particles exist in two internal states — ground (g, marked by $\texttt{tp} = 0$) and excited (e, marked by $\texttt{tp} = -1$) — separated by a mass splitting δ (specified in keV).

Three scattering channels are possible:

(i) $g + g \rightarrow e + e$ (endothermic): requires $\frac{1}{2}\mu v_{\rm rel}^2 > 2\delta$, where $\mu = m_i m_j/(m_i + m_j)$ is the reduced mass. The post-scattering relative speed decreases: $|v'_{\rm rel}| = \sqrt{2(\frac{1}{2}\mu v_{\rm rel}^2 - 2\delta)/\mu}$.

(ii) $e + e \rightarrow g + g$ (exothermic): always kinematically allowed. The relative speed increases: $|v'_{\rm rel}| = \sqrt{2(\frac{1}{2}\mu v_{\rm rel}^2 + 2\delta)/\mu}$.

(iii) $g{+}e$ or $e{+}g$: elastic scattering only, no state transition.

At startup, a fraction `sidm_frac_excited` of DM particles is randomly assigned to the excited state. This initialisation is restart-safe: if excited particles already exist in the restart file, the initialisation is skipped.

5.6 Timestep Constraint

The scattering probability (equation 20) must remain below unity for the Monte Carlo algorithm to be valid. We enforce this by tracking the maximum probability $P_{\rm max}$ at each AMR level during the scattering sweep and imposing a timestep constraint in the Courant condition:

$$\Delta t_{\rm new} \leq \frac{C_{\rm SIDM}}{P_{\rm max}}\,\Delta t_{\rm old}, \tag{34}$$

where $C_{\rm SIDM}$ (default 0.1) is the SIDM Courant number specified by `sidm_courant`. This ensures that the scattering probability remains below $C_{\rm SIDM}$ at the next step.

5.7 OpenMP Thread Safety

The scattering algorithm is parallelised over grids using the cuRAMSES `pthreadLinkedList` OpenMP dispatch, following the pattern established for stellar feedback (Kim 2026). Thread safety requires careful handling of the random number generator (RNG) state. We maintain a per-thread seed array `sidm_seeds(IRandNumSize, 0:nthreads-1)`, initialised from RAMSES's global `localseed` with 100 RNG advances between consecutive threads to decorrelate the streams. After the OpenMP region, `localseed` is updated from the last thread's seed (advanced 100 steps) for deterministic reproducibility across successive calls.

Thread-local diagnostic counters (scatter counts, momentum and energy residuals) are accumulated to shared totals via `!$omp critical` sections.

5.8 Conservation Diagnostics

For each scattering event, the code tracks the cumulative changes in momentum ($\Delta\mathbf{p}$) and kinetic energy ($\Delta E_{\rm k}$), which should be zero for elastic scattering. These are reduced across MPI ranks (`MPI_SUM` for totals, `MPI_MAX` for worst-case single-event residuals) and reported in the simulation log, providing runtime verification of the conservation properties.

5.9 Namelist Configuration

The SIDM module is activated by `sidm=.true.` in `&RUN_PARAMS`. All scattering parameters are specified in a dedicated `&SIDM_PARAMS` block:

Table 3. SIDM namelist parameters (`&SIDM_PARAMS`).

Parameter	Type	Default	Description
`sidm_cross_section`	real	1.0	σ_0/m [cm^2 g^{-1}]
`sidm_type`	string	`constant`	`constant/yukawa/power_law`
`sidm_v0`	real	100.0	Reference velocity [km s^{-1}]
`sidm_power`	real	-4.0	Power-law index n
`sidm_npart_min`	int	2	Min. DM particles per cell
`sidm_courant`	real	0.1	Max. scatter probability
`sidm_angular`	string	`isotropic`	`isotropic/rutherford`
`sidm_epsilon`	real	0.01	Rutherford regularisation ε
`sidm_inelastic`	logical	F	Enable iSIDM
`sidm_delta`	real	0.0	Mass splitting [keV]
`sidm_frac_excited`	real	0.0	Initial excited fraction

```
&SIDM_PARAMS
sidm_cross_section = 1.0
sidm_type = 'constant'
sidm_v0 = 100.0
sidm_power = -4.0
sidm_npart_min = 2
sidm_courant = 0.1
sidm_angular = 'isotropic'
sidm_epsilon = 0.01
sidm_inelastic = .false.
sidm_delta = 0.0
sidm_frac_excited = 0.0
/
```

Table 3 summarises all parameters with their units and defaults.

6 MODIFIED NEWTONIAN DYNAMICS (QUMOND AND AQUAL)

6.1 Physical Motivation

Modified Newtonian Dynamics (Milgrom 1983) postulates that the effective gravitational acceleration deviates from the Newtonian prediction $\mathbf{g}_{\rm N}$ when $|\mathbf{g}_{\rm N}| \lesssim a_0$, where $a_0 \approx 1.2 \times 10^{-8}\,{\rm cm\,s^{-2}}$ is a universal acceleration scale. MOND successfully reproduces observed galaxy rotation curves, the baryonic Tully–Fisher relation, and the radial acceleration relation without invoking dark matter on galactic scales (Famaey & McGaugh 2012; Milgrom 2014; McGaugh et al. 2016).

A key requirement for N-body implementations is that the modified gravitational field be derivable from a potential (i.e. curl-free), so that energy is conserved. The quasi-linear formulation of MOND (QUMOND; Milgrom 2010) achieves this by defining the MOND potential Φ through a modified Poisson equation:

$$\nabla^2\Phi = \nabla\cdot\left[\nu(|\nabla\Phi_{\rm N}|/a_0)\,\nabla\Phi_{\rm N}\right], \tag{35}$$

where $\Phi_{\rm N}$ is the standard Newtonian potential satisfying $\nabla^2\Phi_{\rm N} = 4\pi G\rho$, and $\nu(y)$ is an interpolation function satisfying $\nu \rightarrow 1$ for $y \gg 1$ (Newtonian regime) and $\nu \rightarrow y^{-1/2}$ for $y \ll 1$ (deep-MOND regime).

An alternative, more fundamental formulation is AQUAL (AQUAdratic Lagrangian; **?**), which derives from an action principle. The AQUAL modified Poisson equation is

$$\nabla\cdot\left[\mu(|\nabla\Phi|/a_0)\,\nabla\Phi\right] = 4\pi G\rho, \tag{36}$$

where $\mu(x)$ is the MOND interpolation function satisfying $\mu \to 1$ for $x \gg 1$ and $\mu \to x$ for $x \ll 1$. Unlike QUMOND, the coefficient μ depends on the MOND potential Φ itself, making equation (36) nonlinear. The μ and ν functions are related: for the simple family, $\mu(x) = x/(1+x)$ and $\nu(y) = \frac{1}{2} + \frac{1}{2}\sqrt{1+4/y}$, where $y = x\mu(x)$.

6.2 Interpolation Functions

We implement two standard interpolation functions, selected by `mond_mu_type`:

- **Simple** (`mond_mu_type=1`): corresponding to $\mu(x) = x/(1+x)$, the ν-function is

$$\nu(y) = \frac{1}{2} + \frac{1}{2}\sqrt{1+4/y}. \tag{37}$$

- **Standard** (`mond_mu_type=2`): corresponding to $\mu(x) = x/\sqrt{1+x^2}$, the ν-function is

$$\nu(y) = \sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{1+4/y^2}}. \tag{38}$$

Here $y \equiv |\mathbf{g}_\mathrm{N}|/a_0$ with $\mathbf{g}_\mathrm{N} = -\nabla\Phi_\mathrm{N}$. Both functions are evaluated cell-by-cell from the Newtonian force array $f_i = -\partial\Phi_\mathrm{N}/\partial x_i$ that is already computed by `force_fine`.

For AQUAL (Phase 2), the corresponding μ-functions are:

- **Simple**: $\mu(x) = x/(1+x)$, so $\mu - 1 = -1/(1+x)$.
- **Standard**: $\mu(x) = x/\sqrt{1+x^2}$, so $\mu - 1 = x/\sqrt{1+x^2} - 1$.

Note that $\mu - 1 < 0$ always (since $\mu < 1$), whereas $\nu - 1 > 0$ always (since $\nu > 1$).

6.3 Phase 0: Algebraic Force Correction

The simplest implementation (`mond_type=0`) applies the QUMOND correction algebraically:

$$f_i^\mathrm{MOND} = \nu(|\mathbf{f}|/a_0)\, f_i, \tag{39}$$

where $\mathbf{f}$ is the Newtonian force vector. This is performed in the subroutine `apply_mond_force`, called immediately after `gradient_phi` within `force_fine`.

This approach is computationally inexpensive (a single pass over all cells with no additional Poisson solve) and sufficient for spherically symmetric mass distributions where the curl of $\nu\mathbf{g}_\mathrm{N}$ vanishes by symmetry. However, for general mass distributions the resulting field $\nu\mathbf{g}_\mathrm{N}$ is not curl-free, leading to a non-conservative force and energy conservation violations.

6.4 Phase 1: Full QUMOND with Phantom Density

To guarantee a curl-free MOND gravitational field, the full QUMOND formulation (`mond_type=1`) introduces a *phantom dark matter density* ρ_ph and re-solves the Poisson equation. Expanding equation (35):

$$\nabla^2\Phi = \nabla^2\Phi_\mathrm{N} + \nabla\cdot\big[(\nu-1)\nabla\Phi_\mathrm{N}\big] = 4\pi G\rho + 4\pi G\rho_\mathrm{ph}, \tag{40}$$

so the phantom density is

$$\rho_\mathrm{ph} = \frac{1}{4\pi G}\nabla\cdot\big[(\nu-1)\nabla\Phi_\mathrm{N}\big] = -\frac{1}{4\pi G}\nabla\cdot\big[(\nu-1)\,\mathbf{f}\big], \tag{41}$$

where we used $\nabla\Phi_\mathrm{N} = -\mathbf{f}$. In cosmological mode, $4\pi G$ is replaced by $\frac{3}{2}\Omega_\mathrm{m} a$.

6.4.1 Divergence computation

The divergence in equation (41) is computed using second-order centred differences on the AMR grid:

$$\nabla\cdot\big[(\nu-1)\mathbf{f}\big] \approx \sum_{d=1}^{3} \frac{h_d^+ - h_d^-}{2\,\Delta x}, \tag{42}$$

where $h_d^\pm = [\nu(|\mathbf{f}|/a_0) - 1]\, f_d$ evaluated at the right (+) and left (−) face-neighbours in dimension d. The neighbour cell indices are obtained from the same `ggg/hhh` lookup tables used by `gradient_phi` in `force_fine`, ensuring consistent stencil geometry. At coarse–fine level boundaries where a face-neighbour grid does not exist at the current refinement level (`igridn` = 0), the cell is skipped; this affects only a small fraction of cells near AMR boundaries.

6.4.2 Algorithm

The full QUMOND algorithm within a single AMR level proceeds as:

(i) Solve the standard Poisson equation: $\nabla^2\Phi_\mathrm{N} = 4\pi G\rho$.
(ii) Compute Newtonian forces: $\mathbf{f} = -\nabla\Phi_\mathrm{N}$ via `force_fine`.
(iii) Compute phantom density ρ_ph from equation (41) and add to ρ in-place (`compute_mond_phantom_density`).
(iv) Exchange boundaries for the modified ρ.
(v) Re-solve the Poisson equation: $\nabla^2\Phi = 4\pi G(\rho + \rho_\mathrm{ph})$.
(vi) Compute MOND forces: $\mathbf{f}_\mathrm{MOND} = -\nabla\Phi$ via `force_fine`.

Steps 3–6 are inserted in `amr_step` between the initial `force_fine` call and the particle synchronisation (`synchro_fine`). The algebraic correction (`apply_mond_force`) is automatically disabled when `mond_type`≥ 1.

6.4.3 Thread safety

The phantom density computation is fully thread-safe under OpenMP: each cell's ρ is modified by exactly one thread (the cell index is determined uniquely by the grid–octant pair in the parallel loop), while the neighbour force values $\mathbf{f}$ are accessed read-only. All temporary variables (neighbour indices, $\nu - 1$ values, divergence accumulator) are declared `private`.

6.5 Phase 2: AQUAL Iterative Solver

The AQUAL formulation (`mond_type=2`) solves the nonlinear modified Poisson equation (36) self-consistently via fixed-point iteration. Rewriting equation (36) as

$$\nabla^2\Phi = 4\pi G\rho + \nabla\cdot\big[(\mu-1)\,\mathbf{f}\big], \tag{43}$$

where $\mathbf{f} = -\nabla\Phi$ and $\mu = \mu(|\mathbf{f}|/a_0)$, we define an AQUAL phantom density

$$\rho_\mathrm{ph}^\mathrm{AQUAL} = +\frac{1}{4\pi G}\nabla\cdot\big[(\mu-1)\,\mathbf{f}\big]. \tag{44}$$

Note the sign difference from the QUMOND phantom density (equation 41): the QUMOND term uses $\nu - 1 > 0$ with a minus sign, while the AQUAL term uses $\mu - 1 < 0$ with a plus sign; both yield positive ρ_ph in the deep-MOND regime.

6.5.1 Algorithm

The AQUAL iteration proceeds as follows:

(i) Solve the standard Poisson equation to obtain the Newtonian force $\mathbf{f}^{(0)} = -\nabla\Phi_{\rm N}$.

(ii) For $n = 0, 1, \ldots, n_{\rm iter}$:

(a) Restore ρ to the baryonic density $\rho_{\rm bary}$.

(b) Compute the AQUAL phantom density $\rho_{\rm ph}^{(n)} = +\nabla\cdot[(\mu-1)\mathbf{f}^{(n)}]/(4\pi G)$.

(c) Add $\rho_{\rm ph}^{(n)}$ to ρ and exchange boundaries.

(d) Re-solve the Poisson equation: $\nabla^2\Phi^{(n+1)} = 4\pi G(\rho + \rho_{\rm ph}^{(n)})$.

(e) Compute $\mathbf{f}^{(n+1)} = -\nabla\Phi^{(n+1)}$.

(f) Check convergence: $\max|\mathbf{f}^{(n+1)} - \mathbf{f}^{(n)}|/\max|\mathbf{f}^{(n+1)}| < \epsilon$.

(iii) Return the converged force $\mathbf{f}^{(\rm final)}$.

The convergence criterion ϵ is controlled by `mond_eps` (default 10^{-6}). Convergence is monitored globally across all MPI ranks via `MPI_ALLREDUCE` with the `MPI_MAX` operator.

6.5.2 Implementation

The iteration is implemented in the subroutine `aqual_iterate`, called from `amr_step` when `mond_type=2`. The baryonic density is saved in a temporary array at the start of the iteration and restored at each step to prevent accumulation of phantom density. The existing `compute_mond_phantom_density` routine is reused with a logical flag `is_aqual` that selects between $\nu - 1$ (QUMOND) and $\mu - 1$ (AQUAL), and applies the appropriate sign convention.

Each iteration calls the full multigrid (or FFT) Poisson solver and `force_fine`, which automatically use the GPU-accelerated or FFTW3 code paths when available. The multigrid solver overwrites the $f()$ array internally (using it as scratch for residual/RHS/mask), so `force_fine` must be called after each solve to reconstruct the forces.

6.6 External Field Effect

In MOND, an external gravitational field $\mathbf{g}_{\rm ext}$ breaks the internal dynamics of a subsystem, even if it is uniform (the "external field effect" or EFE; Milgrom 1983; **?**). This is relevant for isolated galaxy simulations embedded in a large-scale tidal field.

We implement the EFE by replacing $\mathbf{f}$ with $\mathbf{f} + \mathbf{g}_{\rm ext}$ in the argument of the interpolation function. The external field is specified via `g_ext_mond(1:3)` in CGS units and converted to code units at each timestep. The EFE modifies each MOND mode:

- **Phase 0**: $f_d^{\rm MOND} = \nu(|\mathbf{f} + \mathbf{g}_{\rm ext}|/a_0)\, f_d + (\nu - 1)\, g_{{\rm ext},d}$.
- **Phase 1**: the divergence uses $(\nu - 1)(f_d + g_{{\rm ext},d})$.
- **Phase 2**: the divergence uses $(\mu - 1)(f_d + g_{{\rm ext},d})$.

When $\mathbf{g}_{\rm ext} = 0$ (the default), all expressions reduce to the standard EFE-free forms. For cosmological periodic boxes the EFE is automatically zero.

Table 4. MOND namelist parameters (`&MOND_PARAMS`).

Parameter	Type	Default	Description
`a0_mond`	real	1.2×10^{-8}	a_0 [cm s^{-2}]
`mond_mu_type`	integer	1	1=simple, 2=standard
`mond_type`	integer	0	0=algebraic, 1=QUMOND, 2=AQU
`n_iter_mond`	integer	10	Max AQUAL iterations (type=2 onl
`mond_eps`	real	10^{-6}	AQUAL convergence threshold
`g_ext_mond`	real(3)	0,0,0	External field [cm s^{-2}] (EFE)

6.7 Cosmological Considerations

In cosmological N-body simulations, the Poisson equation uses the density contrast and the force $\mathbf{f}$ represents the peculiar gravitational acceleration (the perturbation from the homogeneous Hubble flow). The MOND correction is therefore applied to $|\mathbf{f}|$ rather than to the total gravitational acceleration. This "naive cosmological QUMOND" approach is the standard choice for MOND N-body cosmological codes (Lüghausen et al. 2015; Candlish et al. 2015; Nagesh et al. 2021) and is valid in the quasi-static regime relevant for structure formation.

The critical acceleration a_0 is specified in CGS units (cm s^{-2}) and converted to code units at each timestep via

$$a_0^{\rm code} = a_0/(L_{\rm box}/t_{\rm code}^2), \tag{45}$$

where $L_{\rm box}$ and $t_{\rm code}$ are the time-dependent code length and time units returned by the RAMSES `units()` routine. Since these depend on the scale factor a, the code-unit threshold evolves correctly with cosmic time.

6.8 Performance Impact

Phase 0 (`mond_type=0`) adds negligible overhead: a single pass over all cells to multiply $\mathbf{f}$ by ν. Phase 1 (`mond_type=1`) approximately doubles the Poisson solver cost per level, since the multigrid (or FFT) solve and `force_fine` are each called twice. For typical production runs where the Poisson solver accounts for ∼36 per cent of the total step time, the overall cost increase is ∼36 per cent (i.e. the total step time increases from 1.0× to ∼1.36×).

Phase 2 (`mond_type=2`) requires $N_{\rm iter}+1$ Poisson solves and force computations per level. With a typical convergence in 5–10 iterations, the cost is ∼6–11× the baseline Poisson cost. Using the FFTW3 base-level solver (28.9 s per coarse step), Phase 2 with 10 iterations would cost ∼300 s per coarse step. This is acceptable for isolated galaxy simulations with moderate box sizes but prohibitive for large cosmological volumes.

6.9 Namelist Configuration

The MOND module is enabled by setting `use_mond=.true.` in `&RUN_PARAMS` and providing the physical parameters in a dedicated `&MOND_PARAMS` block:

A minimal QUMOND example:

```
&RUN_PARAMS
use_mond=.true.
/

&MOND_PARAMS
```

```
a0_mond=1.2d-8
mond_mu_type=1
mond_type=1
/
```

An AQUAL example with external field:

```
&MOND_PARAMS
a0_mond=1.2d-8
mond_mu_type=1
mond_type=2
n_iter_mond=10
mond_eps=1d-6
g_ext_mond=0d0,0d0,1d-9
/
```

The code validates that `poisson=.true.` when MOND is enabled, that $a_0 > 0$, that `mond_mu_type`$\in \{1, 2\}$, and that `mond_type`$\in \{0, 1, 2\}$. For `mond_type=2`, `n_iter_mond`≥ 1 and `mond_eps` > 0 are required. A warning is issued if `pic=.true.` (dark matter particles present), since MOND is designed to replace dark matter rather than augment it.

7 $F(R)$ HU–SAWICKI GRAVITY

7.1 Theory

$f(R)$ gravity (Sotiriou & Faraoni 2010) replaces the Einstein–Hilbert action with $S = \int d^4x\,\sqrt{-g}\,[R+f(R)]/(16\pi G)$, where $f(R)$ is a nonlinear function of the Ricci scalar. The Hu–Sawicki model (Hu & Sawicki 2007) is defined by

$$f(R) = -m^2\,\frac{c_1(R/m^2)^n}{c_2(R/m^2)^n + 1}, \tag{46}$$

where $m^2 = H_0^2\Omega_{\rm m}$ and the ratio c_1/c_2 is fixed by the requirement $f(R_0) \approx -2\Lambda$. The scalar degree of freedom $f_R \equiv df/dR$ obeys a Klein–Gordon equation that, in the quasi-static limit, reduces to

$$\nabla^2 f_R = -\frac{a^2}{3}\left[R(f_R) - \bar{R}\right] + \frac{a^2}{3}\,8\pi G\,\delta\rho, \tag{47}$$

where $\bar{R}$ is the background Ricci scalar and $\delta\rho$ is the matter overdensity. The Hu–Sawicki inversion $R(f_R) = \bar{R}_0\,[-n\,f_{R0}/f_R]^{1/(n+1)}$ maps the field value back to the curvature. In high-density regions, the effective mass of f_R becomes large, suppressing deviations from GR — the *chameleon screening mechanism*.

7.2 Solver

We solve equation (47) on each AMR level using Newton–Gauss–Seidel (N-GS) relaxation with red–black ordering. The field f_R is stored in the `scalar_gr` array, initialised to the background value $\bar{f}_R(a)$ on the first step and warm-started from the previous step thereafter. The Jacobian of the residual is

$$\left.\frac{\partial F}{\partial f_R}\right|_c = -\frac{6}{\Delta x^2} - \frac{a^2}{3}\frac{dR}{df_R}, \tag{48}$$

which includes both the standard Laplacian stencil and the nonlinear $R(f_R)$ term. The fifth force is

$$\mathbf{F}_5 = -\tfrac{1}{2}\,\nabla f_R, \tag{49}$$

computed via the shared `compute_fifth_force` routine. Convergence is monitored by $\max|F|/\max|S| < \epsilon$ with default $\epsilon = 10^{-6}$.

7.3 Namelist Configuration

The $f(R)$ module is controlled by `use_fR=.true.` in `&RUN_PARAMS` and the dedicated `&FR_PARAMS` block: `fR0` (amplitude, < 0; default -10^{-6}), `fR_n` (Hu–Sawicki index; default 1), `n_iter_fR` (max iterations; default 20), `fR_eps` (convergence threshold; default 10^{-6}).

8 NDGP BRANEWORLD GRAVITY

8.1 Theory

The Dvali–Gabadadze–Porrati (DGP) model (Dvali et al. 2000) embeds our 4D spacetime in a 5D bulk, with a crossover scale r_c below which gravity is 4-dimensional. In the quasi-static limit, the scalar degree of freedom φ (the brane-bending mode) satisfies

$$\nabla^2\varphi + \frac{r_c^2}{3\beta a^2}\left[(\nabla^2\varphi)^2 - (\nabla_i\nabla_j\varphi)^2\right] = \frac{\Omega_{\rm m}\,a}{\beta}\,\delta, \tag{50}$$

where $\beta(a) = 1 + 2Hr_c(1 + \dot{H}/(3H^2))$ and $\delta = \delta\rho/\bar{\rho}$. The nonlinear derivative terms provide *Vainshtein screening*: near massive objects the second-order operators dominate, suppressing the scalar force.

The normal branch (nDGP; $\epsilon = +1$) does not self-accelerate and requires an explicit cosmological constant, but provides a useful testbed for Vainshtein screening. The self-accelerating branch ($\epsilon = -1$) suffers from ghost instabilities but is included for completeness.

8.2 Solver

We solve equation (50) using Newton–Gauss–Seidel relaxation, identical in structure to the $f(R)$ solver but with the additional Vainshtein bilinear term. The diagonal approximation (Li et al. 2013) drops off-diagonal second derivatives ($\partial_i\partial_j\varphi$ with $i \neq j$), which is accurate to the level of the grid resolution. The Jacobian is

$$J = -\frac{6}{\Delta x^2} + \frac{r_c^2}{3\beta a^2}\cdot 2\nabla^2\varphi\cdot\left(-\frac{4}{\Delta x^2}\right). \tag{51}$$

The fifth force is $\mathbf{F}_5 = -[1/(2\beta)]\,\nabla\varphi$, computed via `compute_fifth_force`.

8.3 Namelist Configuration

Controlled by `use_nDGP=.true.` in `&RUN_PARAMS` and `&NDGP_PARAMS`: `omega_rc` ($\Omega_{r_c} = 1/(4r_c^2H_0^2)$; default 0.25), `nDGP_branch` ($+1$ normal, -1 self-accelerating), `n_iter_nDGP` (default 20), `nDGP_eps` (default 10^{-6}).

9 SYMMETRON GRAVITY

9.1 Theory

The Symmetron model (Hinterbichler & Khoury 2010; Hinterbichler et al. 2011) features a scalar field ϕ with a $\mathbb{Z}_2$-symmetric potential

$$V(\phi) = -\frac{\mu^2}{2}\phi^2 + \frac{\lambda}{4}\phi^4 + V_0, \tag{52}$$

and a universal coupling to matter $A(\phi) = 1 + \phi^2/(2M^2)$. The effective potential in a region of density ρ is

$$V_{\rm eff}(\phi) = \frac{1}{2}\Big(\frac{\rho}{\rho_{\rm ssb}} - 1\Big)\frac{\mu^2}{2}\phi^2 + \frac{\lambda}{4}\phi^4, \tag{53}$$

where $\rho_{\rm ssb} = \mu^2 M^2$ is the critical density. When $\rho > \rho_{\rm ssb}$, the effective mass-squared is positive and the field sits at $\phi = 0$ (symmetry restored), suppressing the fifth force. When $\rho < \rho_{\rm ssb}$, symmetry is spontaneously broken and $\phi \neq 0$ generates a fifth force — the *symmetron screening mechanism*.

In terms of the normalised field $\chi = \phi/\phi_0$ and the Compton wavelength L, the quasi-static equation of motion is

$$\nabla^2\chi = \frac{a^2}{2L^2}\left[\Big(\frac{\rho}{\rho_{\rm ssb}} - \frac{a_{\rm ssb}^3}{a^3}\Big)\chi + \chi^3\right]. \tag{54}$$

9.2 Fifth Force

The Symmetron fifth force is *field-dependent*:

$$\mathbf{F}_5 = -6\,\Omega_{\rm m}\,\beta_{\rm s}^2\,\frac{a_{\rm ssb}}{a}\,\chi\,\nabla\chi, \tag{55}$$

which differs from the $f(R)$ and nDGP cases where the force is a simple gradient of the scalar field. A dedicated `compute_fifth_force_symmetron` routine implements the $\chi\,\nabla\chi$ product.

9.3 Solver

The equation is solved by N-GS relaxation. Denoting the right-hand side coefficients as $m_{\rm eff}^2 = (a^2/2L^2)[(\rho/\rho_{\rm ssb}) - (a_{\rm ssb}/a)^3]$ and $\lambda_{\rm eff} = a^2/(2L^2)$, the residual and Jacobian are

$$F = \nabla^2\chi - m_{\rm eff}^2\,\chi - \lambda_{\rm eff}\,\chi^3, \tag{56}$$

$$J = -6/\Delta x^2 - m_{\rm eff}^2 - 3\lambda_{\rm eff}\,\chi^2. \tag{57}$$

The Compton wavelength L is converted from Mpc/h to code units using $L_{\rm code} = L/L_{\rm box}$.

9.4 Namelist Configuration

Controlled by `use_symmetron=.true.` and `&SYMMETRON_PARAMS`: `a_ssb` (SSB scale factor; default 0.5), `beta_symmetron` (coupling; default 1.0), `L_symmetron` (range in Mpc/h; default 1.0), `n_iter_symmetron` (default 20), `symmetron_eps` (default 10^{-6}).

10 DILATON GRAVITY

10.1 Theory

The environmentally dependent Dilaton model (Brax et al. 2010, 2011) employs the Damour–Polyakov mechanism in which the coupling function $A(\phi_D) = 1 + A_2\,\phi_D^2/2$ has a minimum at the cosmological field value. The quasi-static field equation has the same structure as the Symmetron:

$$\nabla^2\phi_D = \frac{a^2}{2L_D^2}\left[\Big(\frac{\rho}{\rho_0} - \frac{a_0^3}{a^3}\Big)\phi_D + \xi_2\,\phi_D^3\right], \tag{58}$$

where L_D is the Dilaton Compton wavelength, a_0 is the transition scale factor, and ξ_2 parameterises the cubic self-interaction (set to unity in our implementation).

10.2 Fifth Force

The Dilaton fifth force is also field-dependent:

$$\mathbf{F}_5 = -2\,\beta_D^2\,\phi_D\,\nabla\phi_D, \tag{59}$$

implemented in a dedicated `compute_fifth_force_dilaton` routine, structurally identical to the Symmetron force routine but with different coupling coefficients.

10.3 Solver and Namelist

The solver is identical in structure to the Symmetron solver with $(a_{\rm ssb}, L_{\rm sym}) \to (a_0, L_D)$. Parameters are set via `use_dilaton=.true.` and `&DILATON_PARAMS`: `beta_dilaton` (default 1.0), `L_dilaton` (Mpc/h; default 1.0), `a0_dilaton` (default 0.5), `n_iter_dilaton` (default 20), `dilaton_eps` (default 10^{-6}).

11 CUBIC GALILEON GRAVITY

11.1 Theory

The cubic Galileon (Nicolis et al. 2009; Deffayet et al. 2009a) is a scalar field theory with second-order equations of motion and a shift symmetry $\phi \to \phi + c$. The Lagrangian is $\mathcal{L} = c_2\,(\partial\phi)^2 + (c_3/\Lambda^3)\,\Box\phi\,(\partial\phi)^2$, and the quasi-static field equation is

$$\nabla^2\phi_G + \frac{c_3}{c_2 H_0^2 a^2}\left[(\nabla^2\phi_G)^2 - (\nabla_i\nabla_j\phi_G)^2\right] = \frac{\Omega_{\rm m}\,a}{\beta_{\rm G}}\,\delta, \tag{60}$$

where $\beta_{\rm G}(a) = c_2/(6\,c_3\,H)$ and H is the Hubble parameter. This has the same Vainshtein operator as the nDGP equation (50), with $r_c^2/(3\beta)$ replaced by $c_3/(c_2\,H_0^2)$.

11.2 Solver

We reuse the nDGP Gauss–Seidel infrastructure with substituted coefficients: $\text{coeff}_G = c_3/(c_2\,a^2)$ and source $= \Omega_{\rm m}\,a\,\delta/\beta_{\rm G}$. The fifth force is $\mathbf{F}_5 = -[1/(2\beta_{\rm G})]\,\nabla\phi_G$, computed via the shared `compute_fifth_force` routine.

11.3 Namelist Configuration

Controlled by `use_galileon=.true.` and `&GALILEON_PARAMS`: `c2_galileon` (kinetic coefficient; default -1.0), `c3_galileon` (cubic coefficient; default 1.0), `n_iter_galileon` (default 20), `galileon_eps` (default 10^{-6}).

12 COUPLED DARK ENERGY

12.1 Theory

Coupled Dark Energy (CDE) models (Amendola 2000, 2004; Pettorino & Baccigalupi 2008) introduce a direct interaction between dark energy and dark matter in the Einstein frame. For a constant coupling $\beta_{\rm cde}$, the effective gravitational constant experienced by dark matter particles is enhanced:

$$G_{\rm eff} = G\,(1 + 2\beta_{\rm cde}^2). \tag{61}$$

Baryons remain uncoupled and feel only the standard gravitational force. For simplicity, our implementation applies the enhancement to all matter (DM + baryons), which is exact when baryons are a subdominant component and provides a useful approximation otherwise.

12.2 Implementation

After the standard gradient of the gravitational potential is computed in `force_fine`, the `apply_coupled_de_force` subroutine multiplies the force array by $(1 + 2\beta_{\rm cde}^2)$:

$$f_d \to (1 + 2\beta_{\rm cde}^2)\, f_d, \quad d = 1, 2, 3. \tag{62}$$

No scalar field is required; the enhancement is a simple post-processing step applied at the same point as the MOND algebraic correction.

12.3 Namelist Configuration

Controlled by `use_coupled_de=.true.` in `&RUN_PARAMS` and `&COUPLED_DE_PARAMS`: `beta_cde` (coupling constant; default 0.1). Setting $\beta_{\rm cde} = 0$ recovers standard gravity. This module can be used simultaneously with any scalar field model or with early dark energy.

13 EARLY DARK ENERGY

13.1 Theory

Early Dark Energy (EDE) models (Doran & Robbers 2006; Poulin et al. 2019; Hill et al. 2020) introduce a transient dark energy component at early times that modifies the expansion history. We implement the Doran–Robbers parameterisation in which the EDE energy density fraction is

$$f_{\rm ede}(a) = \frac{(a_c/a)^{3(1+w_{\rm ede})}}{1 + (a/a_c)^{3(1+w_{\rm ede})}}, \tag{63}$$

where $a_c = 1/(1 + z_{\rm ede})$ is the critical scale factor and $w_{\rm ede}$ is the EDE equation of state. For $a \ll a_c$, $f_{\rm ede} \to 1$, contributing a constant $\Omega_{\rm ede}$ to the Friedmann equation; for $a \gg a_c$, $f_{\rm ede}$ decays as a power law and EDE becomes negligible.

13.2 Implementation

The EDE contribution enters the Friedmann equation through the `dadtau` and `dadt` functions in `init_time.f90`:

$$H^2(a) = H_0^2\left[\frac{\Omega_{\rm m}}{a^3} + \Omega_\Lambda\, f_{\rm DE}(a) + \Omega_k\, a^{-2} + \Omega_{\rm ede}\, f_{\rm ede}(a)\right]. \tag{64}$$

No modification to the force calculation is needed; the effect propagates entirely through the modified expansion history, which changes the conformal time–scale factor mapping and thus the cosmological time stepping.

13.3 Namelist Configuration

Controlled by `use_ede=.true.` in `&RUN_PARAMS` and `&EDE_PARAMS`: `omega_ede` (EDE fraction; default 0.04), `z_ede` (injection redshift; default 3000), `w_ede` (EDE equation of state; default 1/3). Setting $\Omega_{\rm ede} = 0$ recovers the standard ΛCDM expansion. This module can be used simultaneously with any other model.

14 VERIFICATION

We verify each module using cosmological restart tests from a Horizon Run 5 (Lee et al. 2021) snapshot at $a = 0.164$ (output_00005, step 241), run with 12 MPI ranks. The primary diagnostic is the energy conservation error

$$e_{\rm cons} = \frac{|E_{\rm tot} - E_{\rm tot,0}|}{|E_{\rm tot,0}|}, \tag{65}$$

where $E_{\rm tot}$ is the total (kinetic + potential + thermal) energy.

14.1 ΛCDM Recovery

Each module must reproduce the baseline ΛCDM result when nominally enabled but set to produce no physical effect:

- **Neutrino**: `use_neutrino=.true.`, `omega_nu=0`. The code sets $\Omega_\nu/\Omega_{\rm cb} = 0$, so the neutrino factor in equation (4) is identically unity. Result: $e_{\rm cons} = 6.71 \times 10^{-4}$ at step 242, 6.74×10^{-4} at step 243 — bit-identical to the reference run.
- **Dark energy**: `de_perturb=.true.` with $w_0 = -1$, $w_a = 0$. Since $1 + w = 0$, $\alpha = 0$ (equation 16), so $\mathcal{F}_{\rm DE} = 1$. Result: bit-identical to baseline.
- **SIDM**: `sidm=.false.`. The scattering routine is not called. Result: bit-identical to baseline.
- **MOND**: `use_mond=.true.` with a_0 set to an extremely small value ($\sim 10^{-30}\,{\rm cm\,s^{-2}}$), placing all cells deep in the Newtonian regime ($\nu \approx 1$, $\mu \approx 1$). All three modes (`mond_type=0`, 1, and 2) recover the baseline forces. Setting `use_mond=.false.` is bit-identical to the reference run.
- $f(R)$ **Hu–Sawicki**: `use_fR=.true.` with $|f_{R0}| \to 0$ (e.g. $f_{R0} = -10^{-10}$). The chameleon mass becomes extremely large, screening all cells. The scalar field stays near the background value and the fifth force is negligible.
- **nDGP**: `use_nDGP=.true.` with $\Omega_{r_c} \to \infty$. In this limit $\beta \to \infty$ and the fifth force $\propto 1/\beta \to 0$.
- **Symmetron**: `use_symmetron=.true.` with $\beta_{\rm s} \to 0$ (10^{-10}). The fifth force prefactor $\propto \beta_{\rm s}^2$ vanishes.
- **Dilaton**: `use_dilaton=.true.` with $\beta_D \to 0$ (10^{-10}). Identical argument.
- **Galileon**: `use_galileon=.true.` with $c_3 \to 0$ (10^{-10}). The Vainshtein term vanishes and $\beta_{\rm G} \to \infty$, so the fifth force $\propto 1/\beta_{\rm G} \to 0$.
- **Coupled DE**: `use_coupled_de=.true.` with $\beta_{\rm cde} = 0$. $G_{\rm eff} = G(1 + 0) = G$, identical to standard gravity.
- **Early DE**: `use_ede=.true.` with $\Omega_{\rm ede} = 0$. The EDE term in the Friedmann equation vanishes identically, recovering standard ΛCDM expansion.

14.2 SIDM Conservation

For elastic constant-cross-section SIDM with $\sigma/m = 1\,\mathrm{cm}^2\,\mathrm{g}^{-1}$, the cumulative momentum and kinetic energy conservation residuals over 3 coarse steps are:

$$|\Delta\mathbf{p}|/|\mathbf{p}_{\mathrm{tot}}| \sim 10^{-15}, \tag{66}$$

$$|\Delta E_{\mathrm{k}}|/|E_{\mathrm{k,tot}}| \sim 10^{-15}, \tag{67}$$

consistent with machine-precision round-off. The overall energy conservation error is $e_{\mathrm{cons}} = 7.57 \times 10^{-4}$, comparable to the ΛCDM baseline.

14.3 SIDM Cross-Section Scaling

Increasing σ/m from 1 to $10\,\mathrm{cm}^2\,\mathrm{g}^{-1}$ produces a proportional increase in the total number of scattering events (factor ∼10×), confirming the linear dependence of equation (20) on σ/m.

14.4 OpenMP Reproducibility

The deterministic per-thread seed initialisation (Section 5.7) ensures that the total scatter count and per-event diagnostics are identical between OMP = 1 and OMP = 5 runs, verifying that the threading does not introduce spurious stochasticity.

14.5 MOND Phase Consistency

For a spherically symmetric mass distribution, the curl of $\nu\mathbf{g}_{\mathrm{N}}$ vanishes, so Phase 0 (algebraic) and Phase 1 (phantom density) should produce identical forces. We verify this by comparing the force field at a high-density peak region in the test snapshot: the relative difference $|\mathbf{f}_{\mathrm{Phase\,0}} - \mathbf{f}_{\mathrm{Phase\,1}}|/|\mathbf{f}_{\mathrm{Phase\,1}}|$ is at the 10^{-3} level near the peak (where the geometry is approximately spherical) and reaches a few per cent in asymmetric regions, as expected from the non-vanishing curl field that Phase 1 correctly removes.

For Phase 2 (AQUAL), the converged solution should agree with QUMOND in the curl-free limit (spherical symmetry). In general mass distributions the two formulations differ because AQUAL solves for a self-consistent potential while QUMOND uses the Newtonian potential as input. The AQUAL iteration convergence is monitored at each step via the relative force change $\max|\Delta\mathbf{f}|/\max|\mathbf{f}|$.

14.6 Neutrino Power Spectrum Suppression

With $\Omega_\nu = 0.00140$ ($\sum m_\nu = 0.06\,\mathrm{eV}$), the expected small-scale power spectrum suppression is $\Delta P/P \approx -8\,\Omega_\nu/\Omega_{\mathrm{m}} \approx -3.6$ per cent. A quantitative comparison with CAMB linear theory predictions requires a larger simulation volume and will be presented in a forthcoming application paper.

14.7 Dark Energy Power Spectrum Validation

We validate the DE perturbation module by comparing matter power spectra from gravity-only N-body simulations against CAMB linear theory predictions for four cosmological models: ΛCDM ($w_0 = -1, w_a = 0$), wCDM ($w_0 = -0.8, w_a = 0$), CPL thaw ($w_0 = -0.9, w_a = 0.2$), and CPL phantom ($w_0 = -1.1, w_a = -0.2$). All runs use $N = 256^3$ particles in a $L = 500\,h^{-1}\,\mathrm{Mpc}$ box with $c_{\mathrm{s}}^2 = 0$ and matching initial conditions from MUSIC (Hahn & Abel 2011), evolved from $z = 17$ to $z = 0$ with six output redshifts ($z = 5, 2, 1, 0.5, 0.2, 0$). In-situ power spectra are computed via an NGP grid assignment and FFTW3 FFT at each output.

Fig. 1 shows the ratio $P_{\mathrm{sim}}(k)/P_{\mathrm{CAMB}}(k)$ for each model. In the linear regime ($k < 0.1\,h\,\mathrm{Mpc}^{-1}$), all four models yield ratios of ∼0.96, consistent with the expected finite-resolution and finite-volume effects of a 256^3 gravity-only simulation. The wCDM and CPL-thaw models show a 1–2 per cent enhancement relative to ΛCDM, reflecting the additional gravitational potential from DE perturbations captured by the Weyl-potential $R_{\mathrm{DE}}(k, a)$ table. For the phantom model, DE perturbations are disabled (`de_perturb=F`) since the Weyl method requires non-phantom equations of state; the background expansion history alone correctly reproduces the CAMB predictions.

15 CONCLUSIONS

We have presented eleven non-standard cosmological modules for the cuRAMSES AMR code, spanning three broad categories.

Poisson solver extensions: (i) CPL dark energy perturbations via a table-based linear response method using precomputed CAMB Weyl-potential ratios, valid for arbitrary sound speed including the clustering limit $c_{\mathrm{s}}^2 = 0$; (ii) massive neutrino linear response via scale-dependent $R_\nu(k, a)$ corrections.

Dark matter phenomenology: (iii) SIDM Monte Carlo scattering with constant, velocity-dependent, anisotropic, and inelastic cross-sections, with machine-precision conservation; (iv) QUMOND and AQUAL (three modes: algebraic, phantom density, iterative) with external field effect.

Scalar field modified gravity: (v) $f(R)$ Hu–Sawicki with chameleon screening; (vi) nDGP braneworld with Vainshtein screening; (vii) Symmetron with spontaneous symmetry-breaking screening; (viii) Dilaton with Damour–Polyakov environmentally dependent coupling; (ix) cubic Galileon with Vainshtein screening. All five share a common Newton–Gauss–Seidel solver infrastructure with red–black ordering and OpenMP parallelism.

Force and background modifications: (x) coupled dark energy with $G_{\mathrm{eff}} = G(1 + 2\beta_{\mathrm{cde}}^2)$; (xi) early dark energy in the Doran–Robbers parameterisation, modifying the Friedmann equation at high redshift.

Each module is self-contained, controlled by dedicated namelist blocks, and verified to recover the ΛCDM baseline when disabled. The six scalar field gravity models (MOND, $f(R)$, nDGP, Symmetron, Dilaton, Galileon) are mutually exclusive, while coupled DE and early DE can combine freely with any model.

Together with the algorithmic optimizations described in Kim (2026) — k-section domain decomposition, Morton key octree, spatial hash feedback, and hybrid CPU/GPU dispatch — cuRAMSES now provides a comprehensive platform for precision cosmological simulations that span the standard model and its extensions across all major theoretical frameworks: dynamical dark energy, massive neutrinos,

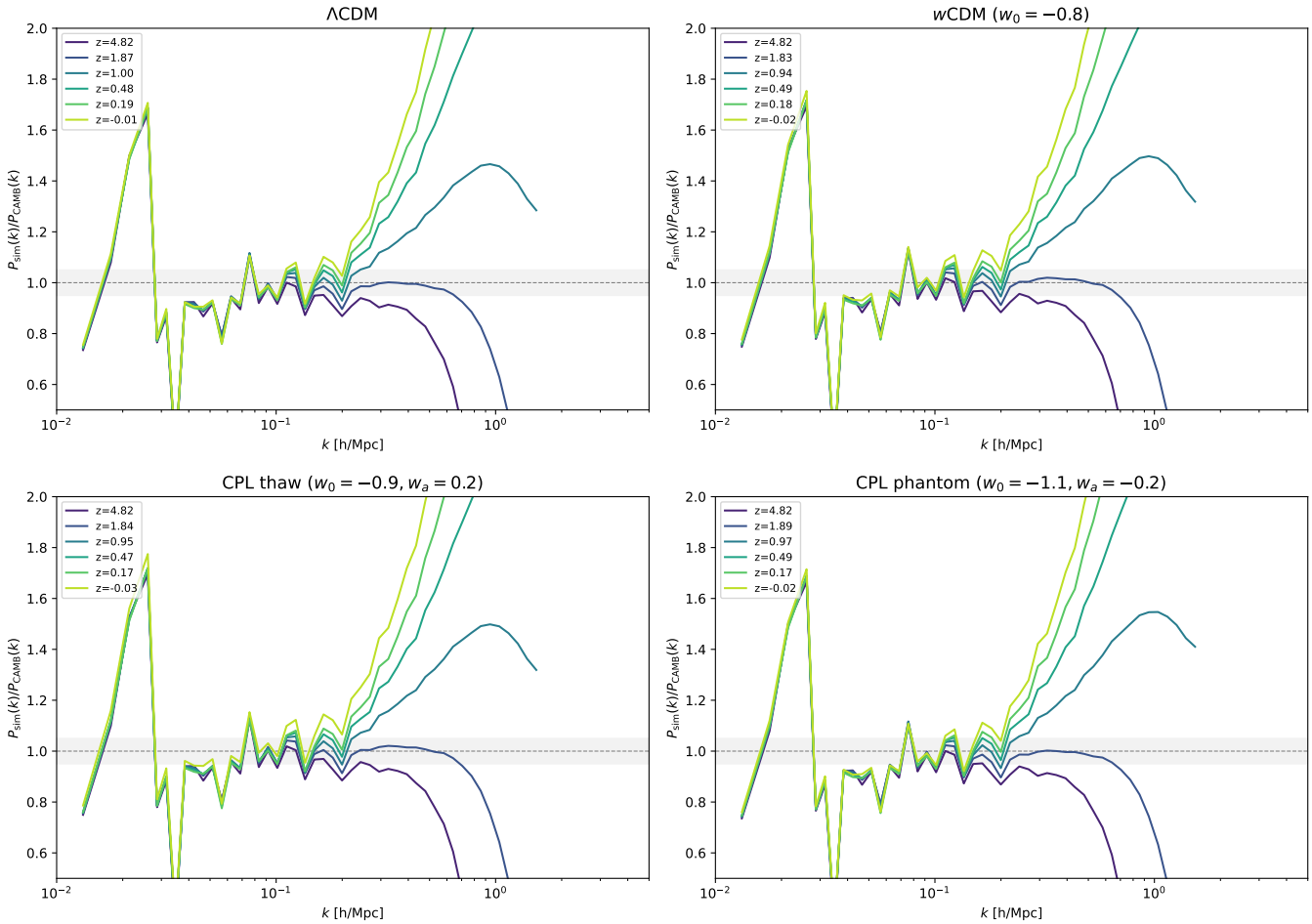

Figure 1. Ratio of N-body matter power spectrum to CAMB linear prediction for four DE models. Each panel shows one model with six output redshifts ($z = 0$–5, colour-coded). The grey band marks ± 5 per cent. All models converge to $P_{\rm sim}/P_{\rm CAMB} \approx 0.96$ in the linear regime, consistent with finite-resolution effects.

self-interacting dark matter, modified gravity, and modified expansion history.

Future work will apply these modules to constrained-realisation simulations targeting DESI and Euclid survey volumes, enabling direct comparison of ΛCDM, w_0w_aCDM, νΛCDM, SIDM, MOND, $f(R)$, nDGP, Symmetron, Dilaton, and Galileon predictions for the halo mass function, galaxy rotation curves, galaxy clustering, and weak lensing observables.

ACKNOWLEDGEMENTS

This work was supported by the Korea Institute for Advanced Study. Computational resources were provided by the KIAS Center for Advanced Computation. The author thanks Romain Teyssier for the public release of the RAMSES code, and Antony Lewis for the CAMB Boltzmann code.

DATA AVAILABILITY

The modified code is available at https://github.com/kjhan0606/cuRAMSES-kjhan. The CAMB table generation scripts (`generate_neutrino_table.py`, `generate_de_table.py`) are included in the `misc/` directory.

REFERENCES

Ali-Haïmoud Y., Bird S., 2013, MNRAS, 428, 3375, doi:10.1093/mnras/sts286, arXiv:1209.0461

Banerjee A., Adhikari S., Dalal N., More S., Kravtsov A., 2020, J. Cosmol. Astropart. Phys., 2020, 024, doi:10.1088/1475-7516/2020/02/024, arXiv:1906.12026

Bean R., Doré O., 2004, Phys. Rev. D, 69, 083503, doi:10.1103/PhysRevD.69.083503, arXiv:astro-ph/0307100

Bird S., Ali-Haïmoud Y., Feng Y., Liu J., 2018, MNRAS, 481, 1486, doi:10.1093/mnras/sty2376, arXiv:1803.09854

Brandbyge J., Hannestad S., Haugbølle T., Thomsen B., 2008, J. Cosmol. Astropart. Phys., 2008, 020, doi:10.1088/1475-7516/2008/08/020, arXiv:0802.3700

Chevallier M., Polarski D., 2001, Int. J. Mod. Phys. D, 10, 213, doi:10.1142/S0218271801000822, arXiv:gr-qc/0009008

DESI Collaboration, 2024, preprint, arXiv:2404.03002

Euclid Collaboration, 2024, A&A, 690, A200, doi:10.1051/0004-6361/202450810, arXiv:2405.13491
Feng J. L., Kaplinghat M., Yu H.-B., 2009, Phys. Rev. Lett., 104, 151301, doi:10.1103/PhysRevLett.104.151301, arXiv:0911.0422
Guillet T., Teyssier R., 2011, J. Comput. Phys., 230, 4756, doi:10.1016/j.jcp.2011.02.044, arXiv:1104.1703
Ivezić Ž., et al., 2019, ApJ, 873, 111, doi:10.3847/1538-4357/ab042c, arXiv:0805.2366
Kahlhoefer F., Schmidt-Hoberg K., Kummer J., Sarkar S., 2014, MNRAS, 452, L54, doi:10.1093/mnrasl/slv088, arXiv:1504.02432
Kaplinghat M., Tulin S., Yu H.-B., 2016, Phys. Rev. Lett., 116, 041302, doi:10.1103/PhysRevLett.116.041302, arXiv:1508.03339
Kim J., 2026, MNRAS, submitted (Paper I)
Knuth D. E., 1997, The Art of Computer Programming, Vol. 3: Sorting and Searching, 2nd edn. Addison-Wesley, Reading, MA
Lattanzi M., Gerbino M., 2017, Frontiers in Physics, 5, 70, doi:10.3389/fphy.2017.00070, arXiv:1712.07109
Lee J., et al., 2021, ApJ, 908, 11, doi:10.3847/1538-4357/abd08b, arXiv:2006.01039
Lesgourgues J., Pastor S., 2006, Phys. Rep., 429, 307, doi:10.1016/j.physrep.2006.04.001, arXiv:astro-ph/0603494
Lewis A., Challinor A., Lasenby A., 2000, ApJ, 538, 473, doi:10.1086/309179, arXiv:astro-ph/9911177
Linder E. V., 2003, Phys. Rev. Lett., 90, 091301, doi:10.1103/PhysRevLett.90.091301, arXiv:astro-ph/0208512
Planck Collaboration, 2020, A&A, 641, A6, doi:10.1051/0004-6361/201833910, arXiv:1807.06209
Robertson A., Massey R., Eke V., 2017, MNRAS, 467, 4719, doi:10.1093/mnras/stx463, arXiv:1612.03906
Rocha M., Peter A. H. G., Bullock J. S., Kaplinghat M., Garrison-Kimmel S., Oñorbe J., Moustakas L. A., 2013, MNRAS, 430, 81, doi:10.1093/mnras/sts514, arXiv:1208.3025
Sapone D., Kunz M., 2010, Phys. Rev. D, 80, 083519, doi:10.1103/PhysRevD.80.083519, arXiv:0909.0007
Spergel D. N., Steinhardt P. J., 2000, Phys. Rev. Lett., 84, 3760, doi:10.1103/PhysRevLett.84.3760, arXiv:astro-ph/9909386
Teyssier R., 2002, A&A, 385, 337, doi:10.1051/0004-6361:20011817, arXiv:astro-ph/0111367
Tucker Smith D., Weiner N., 2001, Phys. Rev. D, 64, 043502, doi:10.1103/PhysRevD.64.043502, arXiv:hep-ph/0101138
Tulin S., Yu H.-B., Zurek K. M., 2013, Phys. Rev. D, 87, 115007, doi:10.1103/PhysRevD.87.115007, arXiv:1302.3898
Tulin S., Yu H.-B., 2018, Phys. Rep., 730, 1, doi:10.1016/j.physrep.2017.11.004, arXiv:1705.02358
Viel M., Haehnelt M. G., Springel V., 2010, J. Cosmol. Astropart. Phys., 2010, 015, doi:10.1088/1475-7516/2010/06/015, arXiv:1003.2422
Vogelsberger M., Zavala J., Cyr-Racine F.-Y., Pfrommer C., Bringmann T., Sigurdson K., 2019, MNRAS, 484, 5437, doi:10.1093/mnras/stz340, arXiv:1808.01843
Hahn O., Abel T., 2011, MNRAS, 415, 2101, doi:10.1111/j.1365-2966.2011.18820.x, arXiv:1103.6031
Milgrom M., 1983, ApJ, 270, 365, doi:10.1086/161130
Milgrom M., 2010, MNRAS, 403, 886, doi:10.1111/j.1365-2966.2009.16184.x, arXiv:0911.5464
Milgrom M., 2014, Scholarpedia, 9, 31410, doi:10.4249/scholarpedia.31410
Famaey B., McGaugh S. S., 2012, Living Rev. Relativ., 15, 10, doi:10.12942/lrr-2012-10, arXiv:1112.3960
McGaugh S. S., Lelli F., Schombert J. M., 2016, Phys. Rev. Lett., 117, 201101, doi:10.1103/PhysRevLett.117.201101, arXiv:1609.05917
Lüghausen F., Famaey B., Kroupa P., 2015, Canadian J. Phys., 93, 232, doi:10.1139/cjp-2014-0168, arXiv:1405.5963
Candlish G. N., Smith R., Fellhauer M., 2015, MNRAS, 446, 1060, doi:10.1093/mnras/stu2158, arXiv:1410.3844
Nagesh S. T., Banik I., Thies I., Kroupa P., Famaey B., Lüghausen F., Wittenburg N., Parziale R., 2021, Canadian J. Phys., 99, 607, doi:10.1139/cjp-2020-0624, arXiv:2009.12155
Hu W., Sawicki I., 2007, Phys. Rev. D, 76, 064004, doi:10.1103/PhysRevD.76.064004, arXiv:0705.1158
Sotiriou T. P., Faraoni V., 2010, Rev. Mod. Phys., 82, 451, doi:10.1103/RevModPhys.82.451, arXiv:0805.1726
Dvali G., Gabadadze G., Porrati M., 2000, Phys. Lett. B, 485, 208, doi:10.1016/S0370-2693(00)00669-9, arXiv:hep-th/0005016
Vainshtein A. I., 1972, Phys. Lett. B, 39, 393, doi:10.1016/0370-2693(72)90147-5
Nicolis A., Rattazzi R., Trincherini E., 2009, Phys. Rev. D, 79, 064036, doi:10.1103/PhysRevD.79.064036, arXiv:0811.2197
Deffayet C., Esposito-Farèse G., Vikman A., 2009, Phys. Rev. D, 79, 084003, doi:10.1103/PhysRevD.79.084003, arXiv:0901.1314
Deffayet C., Deser S., Esposito-Farèse G., 2009, Phys. Rev. D, 80, 064015, doi:10.1103/PhysRevD.80.064015, arXiv:0906.1967
Hinterbichler K., Khoury J., 2010, Phys. Rev. Lett., 104, 231301, doi:10.1103/PhysRevLett.104.231301, arXiv:1001.4525
Hinterbichler K., Khoury J., Levy A., Matas A., 2011, Phys. Rev. D, 84, 103521, doi:10.1103/PhysRevD.84.103521, arXiv:1107.2112
Brax P., van de Bruck C., Davis A.-C., Shaw D., 2010, Phys. Rev. D, 82, 063519, doi:10.1103/PhysRevD.82.063519, arXiv:1005.3735
Brax P., van de Bruck C., Davis A.-C., Li B., Shaw D. J., 2011, Phys. Rev. D, 83, 104026, doi:10.1103/PhysRevD.83.104026, arXiv:1102.3692
Li B., Zhao G.-B., Koyama K., 2012, MNRAS, 421, 3481, doi:10.1111/j.1365-2966.2012.20573.x, arXiv:1111.2602
Li B., Zhao G.-B., Teyssier R., Koyama K., 2013, J. Cosmol. Astropart. Phys., 2013, 023, doi:10.1088/1475-7516/2013/01/023, arXiv:1110.1379
Llinares C., Mota D. F., Winther H. A., 2014, A&A, 562, A78, doi:10.1051/0004-6361/201322412, arXiv:1307.6748
Puchwein E., Baldi M., Springel V., 2013, MNRAS, 436, 348, doi:10.1093/mnras/stt1575, arXiv:1305.2418
Amendola L., 2000, Phys. Rev. D, 62, 043511, doi:10.1103/PhysRevD.62.043511, arXiv:astro-ph/9908023
Amendola L., 2004, Phys. Rev. D, 69, 103524, doi:10.1103/PhysRevD.69.103524, arXiv:astro-ph/0311175
Pettorino V., Baccigalupi C., 2008, Phys. Rev. D, 77, 103003, doi:10.1103/PhysRevD.77.103003, arXiv:0802.0086
Doran M., Robbers G., 2006, J. Cosmol. Astropart. Phys., 2006, 026, doi:10.1088/1475-7516/2006/06/026, arXiv:astro-ph/0601544
Poulin V., Smith T. L., Karwal T., Kamionkowski M., 2019, Phys. Rev. Lett., 122, 221301, doi:10.1103/PhysRevLett.122.221301, arXiv:1811.04083
Hill J. C., McDonough E., Toomey M. W., Alexander S., 2020, Phys. Rev. D, 102, 043507, doi:10.1103/PhysRevD.102.043507, arXiv:2003.07355
Smith T. L., Poulin V., 2020, Phys. Rev. D, 101, 063523, doi:10.1103/PhysRevD.101.063523, arXiv:2009.10740

APPENDIX A: COMPLETE NAMELIST REFERENCE

Table A1 lists all namelist parameters introduced by the non-standard cosmological modules.

Table A1. Namelist parameters for non-standard cosmological modules.

Parameter	Namelist	Type	Default	Description
Global switches (&RUN_PARAMS)				
sidm	&RUN_PARAMS	logical	F	Enable SIDM scattering
use_neutrino	&RUN_PARAMS	logical	F	Enable neutrino linear response
de_perturb	&RUN_PARAMS	logical	F	Enable DE perturbation
use_mond	&RUN_PARAMS	logical	F	Enable QUMOND
use_fR	&RUN_PARAMS	logical	F	Enable $f(R)$ gravity
use_nDGP	&RUN_PARAMS	logical	F	Enable nDGP gravity
use_symmetron	&RUN_PARAMS	logical	F	Enable Symmetron gravity
use_dilaton	&RUN_PARAMS	logical	F	Enable Dilaton gravity
use_galileon	&RUN_PARAMS	logical	F	Enable cubic Galileon gravity
use_coupled_de	&RUN_PARAMS	logical	F	Enable coupled DE
use_ede	&RUN_PARAMS	logical	F	Enable early DE
dump_pk	&RUN_PARAMS	logical	F	Dump matter $P(k)$ at each output
CPL dark energy (&CPL_PARAMS)				
w0	&CPL_PARAMS	real	-1	Present-day EOS w_0
wa	&CPL_PARAMS	real	0	EOS time derivative w_a
cs2_de	&CPL_PARAMS	real	0	DE sound speed c_s^2
de_table	&CPL_PARAMS	string	—	Path to CAMB DE table
Massive neutrinos (&NEUTRINO_PARAMS)				
omega_nu	&NEUTRINO_PARAMS	real	0	Ω_ν
neutrino_table	&NEUTRINO_PARAMS	string	—	Path to CAMB neutrino table
SIDM (&SIDM_PARAMS)				
See Table 3				
MOND (&MOND_PARAMS)				
a0_mond	&MOND_PARAMS	real	1.2×10^{-8}	a_0 [cm s^{-2}]
mond_mu_type	&MOND_PARAMS	integer	1	ν/μ-function: 1=simple, 2=standard
mond_type	&MOND_PARAMS	integer	0	0=algebraic, 1=QUMOND, 2=AQUAL
n_iter_mond	&MOND_PARAMS	integer	10	Max AQUAL iterations
mond_eps	&MOND_PARAMS	real	10^{-6}	AQUAL convergence threshold
g_ext_mond	&MOND_PARAMS	real(3)	0,0,0	External field (EFE) [cm s^{-2}]
$f(R)$ Hu–Sawicki (&FR_PARAMS)				
fR0	&FR_PARAMS	real	-10^{-6}	$f_{R,0}$ amplitude (< 0)
fR_n	&FR_PARAMS	integer	1	Hu–Sawicki index
n_iter_fR	&FR_PARAMS	integer	20	Max N-GS iterations
fR_eps	&FR_PARAMS	real	10^{-6}	Convergence threshold
nDGP (&NDGP_PARAMS)				
omega_rc	&NDGP_PARAMS	real	0.25	Ω_{r_c}
nDGP_branch	&NDGP_PARAMS	integer	$+1$	$+1$=normal, -1=self-accel.
n_iter_nDGP	&NDGP_PARAMS	integer	20	Max N-GS iterations
nDGP_eps	&NDGP_PARAMS	real	10^{-6}	Convergence threshold
Symmetron (&SYMMETRON_PARAMS)				
a_ssb	&SYMMETRON_PARAMS	real	0.5	SSB scale factor
beta_symmetron	&SYMMETRON_PARAMS	real	1.0	Coupling strength
L_symmetron	&SYMMETRON_PARAMS	real	1.0	Range [Mpc/h]
n_iter_symmetron	&SYMMETRON_PARAMS	integer	20	Max iterations
symmetron_eps	&SYMMETRON_PARAMS	real	10^{-6}	Convergence threshold
Dilaton (&DILATON_PARAMS)				
beta_dilaton	&DILATON_PARAMS	real	1.0	Coupling strength
L_dilaton	&DILATON_PARAMS	real	1.0	Range [Mpc/h]
a0_dilaton	&DILATON_PARAMS	real	0.5	Transition scale factor
n_iter_dilaton	&DILATON_PARAMS	integer	20	Max iterations
dilaton_eps	&DILATON_PARAMS	real	10^{-6}	Convergence threshold
Galileon (&GALILEON_PARAMS)				
c2_galileon	&GALILEON_PARAMS	real	-1.0	Kinetic coefficient
c3_galileon	&GALILEON_PARAMS	real	1.0	Cubic coefficient
n_iter_galileon	&GALILEON_PARAMS	integer	20	Max iterations
galileon_eps	&GALILEON_PARAMS	real	10^{-6}	Convergence threshold
Coupled DE (&COUPLED_DE_PARAMS)				
beta_cde	&COUPLED_DE_PARAMS	real	0.1	DM–DE coupling
Early DE (&EDE_PARAMS)				
omega_ede	&EDE_PARAMS	real	0.04	EDE fraction
z_ede	&EDE_PARAMS	real	3000	Injection redshift
w_ede	&EDE_PARAMS	real	1/3	EDE equation of state

APPENDIX B: CAMB TABLE FORMAT

Both the neutrino and dark energy response tables use the same ASCII format:

```
# Header comments (lines starting with #)
nk na
k_1 k_2 ... k_nk       [h/Mpc]
a_1 a_2 ... a_na
R(1,1) R(2,1) ... R(nk,1)
R(1,2) R(2,2) ... R(nk,2)
...
R(1,na) R(2,na) ... R(nk,na)
```

where $R(i,j) = R_X(k_i, a_j)$ is the response function. The k-values are in units of $h\,\mathrm{Mpc}^{-1}$ and the a-values are dimensionless scale factors. Both are stored in log-spacing but written in linear values; the code takes $\log_{10}$ internally for interpolation.

The generation scripts `generate_neutrino_table.py` and `generate_de_table.py` (in `misc/`) produce tables in this format using CAMB v1.6+. The DE table uses the Weyl-potential method (Section 3.2) to extract R_{DE} in the Newtonian gauge. Typical generation time is $\sim$30 s per table.